

1)

When using  $N = 40$  observations to estimate the model

$$y = \beta_1 + \beta_2 x + \beta_3 z + e$$

you obtain  $SSE = 979.830$  and  $s_y = 13.45222$ . Find

- (a)  $R^2$
- (b) The value of the  $F$ -statistic for testing  $H_0: \beta_2 = \beta_3 = 0$  (Do you reject or fail to reject  $H_0$ ?)

2)

Consider the model

$$y = \beta_1 + x_2 \beta_2 + x_3 \beta_3 + e$$

and suppose that application of least squares to 20 observations on these variables yields the following results ( $\widehat{\text{cov}(b)}$  denotes the estimated covariance matrix):

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 0.96587 \\ 0.69914 \\ 1.7769 \end{bmatrix}, \quad \widehat{\text{cov}(b)} = \begin{bmatrix} 0.21812 & 0.019195 & -0.050301 \\ 0.019195 & 0.048526 & -0.031223 \\ -0.050301 & -0.031223 & 0.037120 \end{bmatrix}$$

$$\hat{\sigma}^2 = 2.5193 \quad R^2 = 0.9466$$

- (a) Find the total variation, unexplained variation, and explained variation for this model.
- (b) Find 95% interval estimates for  $\beta_2$  and  $\beta_3$ .
- (c) Use a  $t$ -test to test the hypothesis  $H_0: \beta_2 \geq 1$  against the alternative  $H_1: \beta_2 < 1$ .
- (d) Use your answers in part (a) to test the joint hypothesis  $H_0: \beta_2 = 0, \beta_3 = 0$ .
- (e) Test the hypothesis  $H_0: 2\beta_2 = \beta_3$ .

3)

Consider the wage equation

$$\ln(WAGE) = \beta_1 + \beta_2 EDUC + \beta_3 EDUC^2 + \beta_4 EXPER + \beta_5 EXPER^2 \\ + \beta_6 (EDUC \times EXPER) + \beta_7 HRSWK + e$$

where the explanatory variables are years of education, years of experience and hours worked per week. Estimation results for this equation, and for modified versions of it obtained by dropping some of the variables, are displayed in Table 6.4. These results are from the 1000 observations in the file *cps4c\_small.dat*.

- (a) Using an approximate 5% critical value of  $t_c = 2$ , what coefficient estimates are not significantly different from zero?
- (b) What restriction on the coefficients of Eqn (A) gives Eqn (B)? Use an  $F$ -test to test this restriction. Show how the same result can be obtained using a  $t$ -test.
- (c) What restrictions on the coefficients of Eqn (A) give Eqn (C)? Use an  $F$ -test to test these restrictions. What question would you be trying to answer by performing this test?
- (d) What restrictions on the coefficients of Eqn (B) give Eqn (D)? Use an  $F$ -test to test these restrictions. What question would you be trying to answer by performing this test?
- (e) What restrictions on the coefficients of Eqn (A) give Eqn (E)? Use an  $F$ -test to test these restrictions. What question would you be trying to answer by performing this test?
- (f) Based on your answers to parts (a) to (e), which model would you prefer? Why?
- (g) Compute the missing AIC value for Eqn (D) and the missing SC value for Eqn (A). Which model is favored by the AIC? Which model is favored by the SC?

**Table 6.4** Wage Equation Estimates for Exercises 6.4 and 6.5

| Variable                   | Coefficient Estimates and (Standard Errors) |                         |                      |                         |                         |
|----------------------------|---------------------------------------------|-------------------------|----------------------|-------------------------|-------------------------|
|                            | Eqn (A)                                     | Eqn (B)                 | Eqn (C)              | Eqn (D)                 | Eqn (E)                 |
| <i>C</i>                   | 1.055<br>(0.266)                            | 1.252<br>(0.190)        | 1.573<br>(0.188)     | 1.917<br>(0.080)        | 0.904<br>(0.096)        |
| <i>EDUC</i>                | 0.0498<br>(0.0397)                          | 0.0289<br>(0.0344)      | 0.0366<br>(0.0350)   |                         | 0.1006<br>(0.0063)      |
| <i>EDUC</i> <sup>2</sup>   | 0.00319<br>(0.00169)                        | 0.00352<br>(0.00166)    | 0.00293<br>(0.00170) |                         |                         |
| <i>EXPER</i>               | 0.0373<br>(0.0081)                          | 0.0303<br>(0.0048)      |                      | 0.0279<br>(0.0054)      | 0.0295<br>(0.0048)      |
| <i>EXPER</i> <sup>2</sup>  | −0.000485<br>(0.000090)                     | −0.000456<br>(0.000086) |                      | −0.000470<br>(0.000096) | −0.000440<br>(0.000086) |
| <i>EXPER</i> × <i>EDUC</i> | −0.000510<br>(0.000482)                     |                         |                      |                         |                         |
| <i>HRSWK</i>               | 0.01145<br>(0.00137)                        | 0.01156<br>(0.00137)    | 0.01345<br>(0.00136) | 0.01524<br>(0.00151)    | 0.01188<br>(0.00136)    |
| <i>SSE</i>                 | 222.4166                                    | 222.6674                | 233.8317             | 280.5061                | 223.6716                |
| <i>AIC</i>                 | −1.489                                      | −1.490                  | −1.445               |                         | −1.488                  |
| <i>SC</i>                  |                                             | −1.461                  | −1.426               | −1.244                  | −1.463                  |

4)

Consider the wage equation

$$\ln(WAGE) = \beta_1 + \beta_2 EDUC + \beta_3 EDUC^2 + \beta_4 EXPER + \beta_5 EXPER^2 + \beta_6 HRSWK + e$$

- Suppose you wish to test the hypothesis that a year of education has the same effect on  $\ln(WAGE)$  as a year of experience. What null and alternative hypotheses would you set up?
- What is the restricted model, assuming that the null hypothesis is true?
- Given that the sum of squared errors from the restricted model is  $SSE_R = 254.1726$ , test the hypothesis in (a). (For  $SSE_U$  use the relevant value from Table 6.4. The sample size is  $N = 1,000$ .)

**HW**

5)

Consider the following aggregate production function for the U.S. manufacturing sector:

$$Y = \alpha K^{\beta_2} L^{\beta_3} E^{\beta_4} M^{\beta_5} \exp\{e\}$$

where  $Y$  is gross output,  $K$  is capital,  $L$  is labor,  $E$  is energy, and  $M$  denotes other intermediate materials. The data underlying these variables are given in index form in the file *manuf.dat*.

In Exercise 5.25 we expressed the model

$$Y = \alpha K^{\beta_2} L^{\beta_3} E^{\beta_4} M^{\beta_5} \exp\{e\}$$

in terms of logarithms and estimated it using data in the file *manuf.dat*. Use the data and results from Exercise 5.25 to test the following hypotheses:

- (a)  $H_0 : \beta_2 = 0$  against  $H_1 : \beta_2 \neq 0$ .
- (b)  $H_0 : \beta_2 = 0, \beta_3 = 0$  against  $H_1 : \beta_2 \neq 0$  and/or  $\beta_3 \neq 0$ .
- (c)  $H_0 : \beta_2 = 0, \beta_4 = 0$  against  $H_1 : \beta_2 \neq 0$  and/or  $\beta_4 \neq 0$ .
- (d)  $H_0 : \beta_2 = 0, \beta_3 = 0, \beta_4 = 0$  against  $H_1 : \beta_2 \neq 0$  and/or  $\beta_3 \neq 0$  and/or  $\beta_4 \neq 0$ .
- (e)  $H_0 : \beta_2 + \beta_3 + \beta_4 + \beta_5 = 1$  against  $H_1 : \beta_2 + \beta_3 + \beta_4 + \beta_5 \neq 1$ .
- (f) Analyze the impact of collinearity on this model.

6)

The file *stockton4.dat* contains data on 1500 houses sold in Stockton, California, during 1996–1998. Variable descriptions are in the file *stockton4.def*.

(a) Estimate the following model and report the results:

$$SPRICE = \beta_1 + \beta_2 LIVAREA + \beta_3 AGE + \beta_4 BEDS + \beta_5 BATHS + e$$

- (b) Xiaohui wants to buy a house. She is considering two that have the same living area, the same number of bathrooms, and the same number of bedrooms. One is two years old and the other is ten years old. What price difference can she expect between the two houses? What is a 95% interval estimate for this difference?
- (c) Wanling's house has a living area of 2000 square feet. She is planning to extend her living room by 200 square feet. What is the expected increase in price she will get from this extension? Test as an alternative hypothesis that the increase in price will be at least \$20,000. Use  $\alpha = 0.05$ .
- (d) Xueyan's house has a living area of 1800 square feet. She is planning to add another bedroom of size 200 square feet. What is the expected increase in price she will get from this extension? Find a 95% interval estimate for the expected price increase.

7)

The file *stockton4.dat* contains data on 1,500 houses sold in Stockton, CA during 1996–1998. Variable descriptions are in the file *stockton4.def*.

(a) Estimate the following model

$$\begin{aligned} \ln(SPICE) = & \beta_1 + \beta_2 LIVAREA + \beta_3 LIVAREA^2 + \beta_4 AGE + \beta_5 AGE^2 \\ & + \beta_6 BEDS + \beta_7 (LIVAREA \times BEDS) + \beta_8 (LIVAREA^2 \times BEDS) \\ & + \beta_9 (AGE \times BEDS) + \beta_{10} (AGE^2 \times BEDS) + e \end{aligned}$$

Report the estimated relationship between  $\ln(SPICE)$ , *LIVAREA* and *AGE* for two-, three- and four-bedroom houses.

- (b) Test the null hypothesis  $H_0: \beta_6 = 0, \beta_8 = 0, \beta_9 = 0, \beta_{10} = 0$ . Use  $\alpha = 0.05$ .
- (c) Estimate the model implied by the test result in (b). Report the estimated relationship between  $\ln(SPICE)$ , *LIVAREA* and *AGE* for two-, three- and four-bedroom houses.
- (d) Which of the two models in parts (a) and (c) is favored by, (1) the AIC? (2) the SC?